


Scoreboard
Internal SATA Hard Drives

BRAND MODEL	Maxtor DiamondMax 116H500F0	Seagate Barracuda 7200.9 ST3500641AS	Western Digital Caviar SE16 WD5000AAKS	Western Digital RE2 WD5000YS	Seagate Barracuda 7200.9 ST3750640AS	Hitachi Deskstar 7K1000
						
Price (Rs)	6,700	7,200	7,500	9,800	13,200	21,000
+	+ Value for money	+ Value for money	+ Fast transfer speeds	+ Good performance	+ Good performance	+ Great performance
-	- None in particular	- Average performance	- None in particular	- Expensive	- A bit expensive	- Very expensive
Final Score (Out of 100)	77.55	69.81	78.31	64.11	73.60	72.70
Performance (Out of 70)	50.55	44.68	54.19	52.12	53.04	55.47
Cost/Capacity Index (Out of 30)	27.00	25.13	24.12	18.46	20.56	17.23
Features						
Formatted Capacity (GB)	500	500	500	500	750	1000
Actual Capacity (GB)	465.76	465.76	465.76	465.76	698.64	931.51
Interface Type	SATA II	SATA II	SATA II	SATA II	SATA II	SATA II
Number of Platters	4	4	4	4	4	4
Number of Recording Surfaces	8	8	8	8	8	8
Rotational Speed (RPM)	7200	7200	7200	7200	7200	7200
Buffer Memory (MB)	16	16	16	16	16	32
Performance						
Synthetic Tests						
HD Tach RW 3.0.1.0						
Average Read (MB/s)	59.3	52	71.7	61.4	65.7	72.7
Average Write (MB/s)	52.2	47.7	65.8	54.5	59.5	67.2
CPU Utilisation (%)	2	4	4	2	3	3
Random Access Time (ms)	13.9	13.6	13.7	13.1	14.3	14
Read Burst Speed (MB/s)	235.9	237.7	199.5	191	248.7	197.8
H2Bench						
Interface Speed (50%): Sequential Transfer (MB/s)	161.2	157.5	137.7	117.2	189.3	134.4
Repetitive Sequential Read "Core Test" (MB/s)	183.9	183.4	155.7	145.6	206.3	187.5
Random Access Read (ms)	14.57	13.84	13.37	13.34	13.65	13.88
Random Access Write (ms)	7.51	9.29	6.88	7.16	9.46	6.71
PCMark05						
HDD XP Startup (MB/s)	7.684	5.894	8.978	9.274	8.871	9.571
HDD Application Loading (MB/s)	6.403	6.669	7.618	7.883	6.88	8.401
HDD Virus Scan (MB/s)	127.535	Fail	93.781	90.58	109.786	126.387
SiSoft Sandra Pro Business 2007						
Drive Index (MB/s)	56	53	71	62	64	71
Sequential Read (MB/s)	66	61	82	70	75	83
Random Read (MB/s)	41	42	53	49	46	55
Sequential Write (MB/s)	64	60	80	69	74	77
Random Write (MB/s)	46	42	60	56	49	54
Access Time (ms)	9	7	7	6	8	6
Real-World Tests						
Assorted 4 GB File Write (sec)	73.4	84.4	65.6	71	71	68.2
Assorted 4 GB File Read (sec)	72.4	84.9	73	73	77.2	77.9
Assorted 4 GB Internal Copy (sec)	148.7	164.7	159.2	166.1	137.7	152.7
Sequential 4 GB File Write (sec)	64.1	72.3	56.7	59.5	58.5	55.3
Sequential 4 GB File Read (sec)	63	69.3	57	60.3	57.5	57.7
Sequential 4 GB Internal Copy (sec)	133.4	157.8	164.3	171.9	128.5	143.4
Photoshop CS2 Startup Time (sec)	11.2	12.7	13.2	11.4	11.3	9.5
Image Load (550 MB file) (sec)	44.8	46.6	34.5	37.5	41	38.9
Image Load (1 GB file) (sec)	100	99.1	81.8	84.2	97.2	94.3
Farcry Level Load Time (sec)	30	29.1	26.1	27.8	28.7	26.9
Price Per MB (Paise)	1.40	1.51	1.57	2.05	1.85	2.20